GeoinnovationLab+

# Ukraine Reconstruction Geodata Readiness

## Pilot Study Brief | 10-12 weeks

**Core question:** How current, complete, spatially detailed and interoperable are available geospatial datasets on war-related damage in Ukraine - and how can they support reconstruction planning?

**Background.** Ukraine's reconstruction requires reliable and spatially explicit information on damaged buildings, infrastructure and environmental assets. Relevant information already exists across Ukrainian government systems, international organisations, research projects, satellite-derived products, volunteered geographic information and verified open-source investigations. Yet these sources differ substantially in spatial detail, temporal currency, access conditions, damage classifications and verification status.

**The opportunity.** The pilot does not aim to create another national damage map. Instead, it will assess the reconstruction-readiness of existing geodata and demonstrate how selected sources can be integrated into a transparent, reproducible GIS workflow for a small number of test areas.

## Objectives and research questions

- Identify and classify official, open, volunteered and research-based geospatial sources related to damage and reconstruction.
- Distinguish observation date, reporting date, publication date and last update for each source.
- Assess spatial granularity: national, oblast, hromada, settlement, address, building footprint or event point.
- Compare damage classifications, asset typologies, provenance, uncertainty and verification approaches.
- Test the integration of selected datasets in one or two case-study areas.
- Explore whether recent open Earth observation data can help flag outdated or incomplete information.
- Translate the findings into practical recommendations for reconstruction-oriented users in government, engineering, construction, utilities, finance and insurance.

## Proposed work plan

**WP1 - Data inventory & freshness audit** - Catalogue the relevant sources and assess access, licence, coverage, update cycle, geometry, asset types, damage classes, confidence and limitations.

**WP2 - Common data model** - Define a simple harmonised schema for asset ID, geometry, asset type, damage class, damage/observation dates, source, confidence, reconstruction status and provenance.

**WP3 - Case study** - Integrate and compare selected sources in a well-documented area such as Irpin/Bucha/Hostomel and optionally a second municipality with current reconstruction activity.

**WP4 - EO proof of concept** - Evaluate whether recent Sentinel-1/Sentinel-2 observations and existing open-source methods can support change screening or freshness checks. A full operational damage-detection system is not expected.

**WP5 - Demonstrator** - Create a small QGIS, web-map or dashboard demonstrator showing known damage, source, age of observation, confidence, reconstruction/project status and information gaps.

## Deliverables

- Structured Ukraine war-damage geodata catalogue with a data-freshness assessment
- Source/quality matrix comparing spatial detail, currency, accessibility and reconstruction fitness
- Harmonised case-study dataset in an open geospatial format
- Reproducible QGIS/Python processing workflow
- Small interactive demonstrator or map prototype
- 10-15 page final report and a concise Demo Day presentation

## Candidate profile

**Advanced MSc student or recent graduate in Geoinformatics, GIS, Earth Observation, Remote Sensing or a related field.** Expected skills include confident GIS handling, spatial data integration and metadata assessment; Python, Earth Engine, web mapping or remote sensing experience is an advantage.

## Time frame

**10-12 weeks.** The project is intentionally scoped as a focused innovation study rather than a comprehensive national inventory. The first two weeks should deliver the source catalogue and final case-study design; the final two to three weeks should focus on the prototype, synthesis and presentation.

## Responsible data use

- The study is reconstruction-oriented, not operational conflict intelligence.

- It should use legitimately public data or sources made available under appropriate access conditions.
- It should not collect, infer or publish new operationally sensitive information on military positions or other sensitive current targets.
- No conflict-sensitive OpenStreetMap edits should be made as part of the project.
- Where current critical-infrastructure information is relevant, publication should be limited to information that is already officially public and clearly necessary for the reconstruction use case.

## Why this matters for future partners

**For construction, engineering, energy, utilities, planning, finance and insurance partners, the value lies less in another map than in a transparent answer to three questions:** What is known? How recent and reliable is it? Where are the gaps before a reconstruction decision can be made?

## Selected data landscape - starting points

| Type | Source | What it offers | Relevance / caveat |
|---|---|---|---|
| Official damage & recovery | Registry of Damaged and Destroyed Property (RDDP/RPZM) | Official property/claim/inspection structure; CSV resource schema; updated 5 May 2026. | Official reference; access and geospatial granularity need to be tested. |
| Official reconstruction | DREAM | Public recovery and investment project pipeline; updated national SPP on 16 Sep 2026. | Strong for linking damage/recovery needs to planned projects; not itself a damage inventory. |
| Community/municipal GIS | ONOVA GIS HUB | Object-level documentation of destruction, photos, surveys, restoration status and estimated cost; >25,000 objects reported by Esri case study. | Very relevant operational model; bulk/open access appears limited by access levels. |
| National damage estimates | KSE Institute / Russia Will Pay | Direct-damage estimates based on government, statistical and open data; latest published estimate covers Feb 2026. | Benchmark and sector totals; not primarily an open building-level geodataset. |
| International baseline | RDNA5 | Joint Government of Ukraine / World Bank / EU / UN assessment through 31 Dec 2025. | Authoritative benchmark for damage, losses and reconstruction needs; mostly aggregated. |
| Satellite damage labels | UNITAR/UNOSAT | Building-level assessments with downloadable SHP/GDB for selected AOIs. | Excellent benchmark/training/validation source, but much public coverage dates from 2022. |
| Copernicus emergency mapping | CEMS Rapid & Risk/Recovery Mapping | Activation-based vector/raster products, APIs, metadata and damage grading. | Useful where relevant activations exist; not continuous national coverage. |
| Open research | ETH/UZH Sentinel-1 damage mapping | Open method, code, dashboard and data; building-level probabilities based on Sentinel-1 + Overture. | Best current open methodological starting point; published results extend to Feb 2024. |
| Volunteered baseline | OpenStreetMap / Geofabrik | Frequently updated roads, buildings, POIs and infrastructure baseline. | Excellent reference layer; not a systematic damage register. Conflict-mapping safeguards apply. |
| Open building baseline | Overture Maps | Large-scale building footprints with regular releases. | Useful for consistent national building geometry; no damage status. |
| Local high-resolution studies | RebuildUA | Drone/VHR-based local damage | Strong case-study evidence; |

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|---|---|---|---|
| | | assessments and reconstruction reports for selected municipalities. | coverage and raw-data openness vary. |
| Verified OSINT | Centre for Information Resilience - Eyes on Russia | Geolocated and verified incidents, including civilian infrastructure damage and environmental harm. | Good corroboration/validation source; explicitly non-exhaustive. |
| Environmental thematic map | Greenpeace CEE + Ecoaction | Map launched in 2023 with 30 selected severe environmental-damage cases from ~900 collected cases, satellite-confirmed by Greenpeace. | Useful thematic historical layer; selective and not a current national inventory. |
| Official environmental damage | EcoThreat / EcoZagroza | Official Ministry resource with georeferenced environmental-damage reporting and continuously updated damage statistics; historical functionality currently only partially restored. | Potentially important official environmental layer; current data export/access should be tested. |
| War-risk analytics | War Risks in Ukraine Data Platform | Government-backed incident/risk platform for investors, insurers, IFIs and governments; access by request. | Potentially highly relevant for partner-facing risk analysis; not open access. |

## Selected source links

- Registry of Damaged and Destroyed Property - data.gov.ua
- DREAM public reconstruction platform
- ONOVA GIS HUB - Esri case study
- KSE Institute - direct damage estimate, August 2026
- RDNA5 - UNDP publication page
- UNOSAT - Irpin building damage assessment
- Copernicus EMS On Demand Mapping
- Open-source Sentinel-1 damage mapping paper
- OpenStreetMap Ukraine extracts - Geofabrik
- Overture Maps release information
- RebuildUA - Bucha example
- Centre for Information Resilience - Eyes on Russia map
- Greenpeace/Ecoaction environmental damage map announcement
- EcoThreat / EcoZagroza
- War Risks in Ukraine Data Platform
- HOT - Mapping in Conflict guidance

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